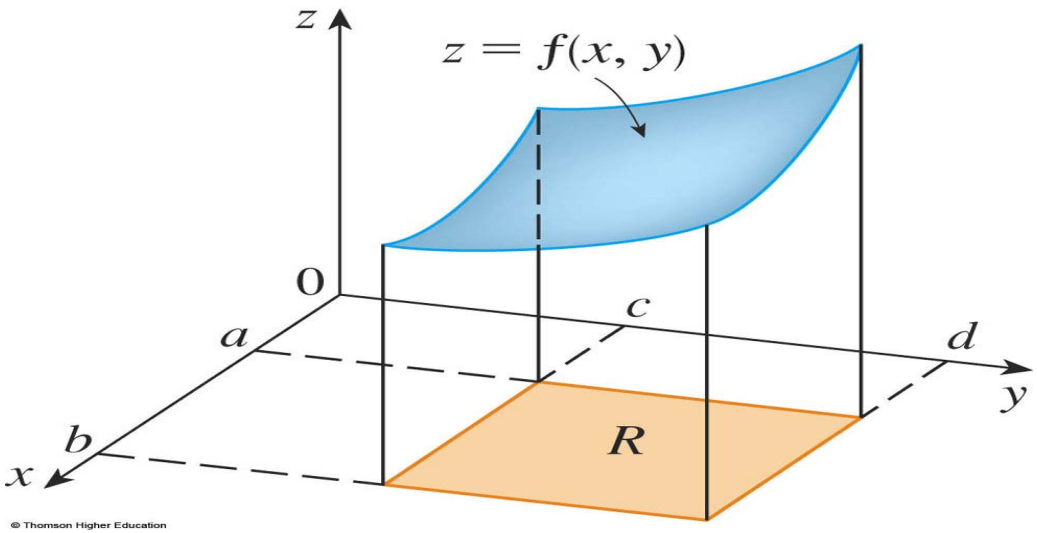
• Double Integrals over Rectangles

We begin with discussing the double integral $\iint_R f(x,y)\,dA$ of a function $f(x,y)$ over a closed rectangle R in $\mathbb{R}^2$, where

$$R = [a,b] \times [c,d] = \{(x,y) \in \mathbb{R}^2 : a \le x \le b,\ c \le y \le d\}.$$



We partition both intervals $[a, b]$ and $[c, d]$ into:

$$a = x_0 < x_1 < x_2 < \ldots < x_{i-1} < x_i < \ldots < x_m = b,$$
$$c = y_0 < y_1 < y_2 < \ldots < y_{j-1} < y_j < \ldots < y_n = d.$$

Then R are partitioned into mn sub-rectangles:

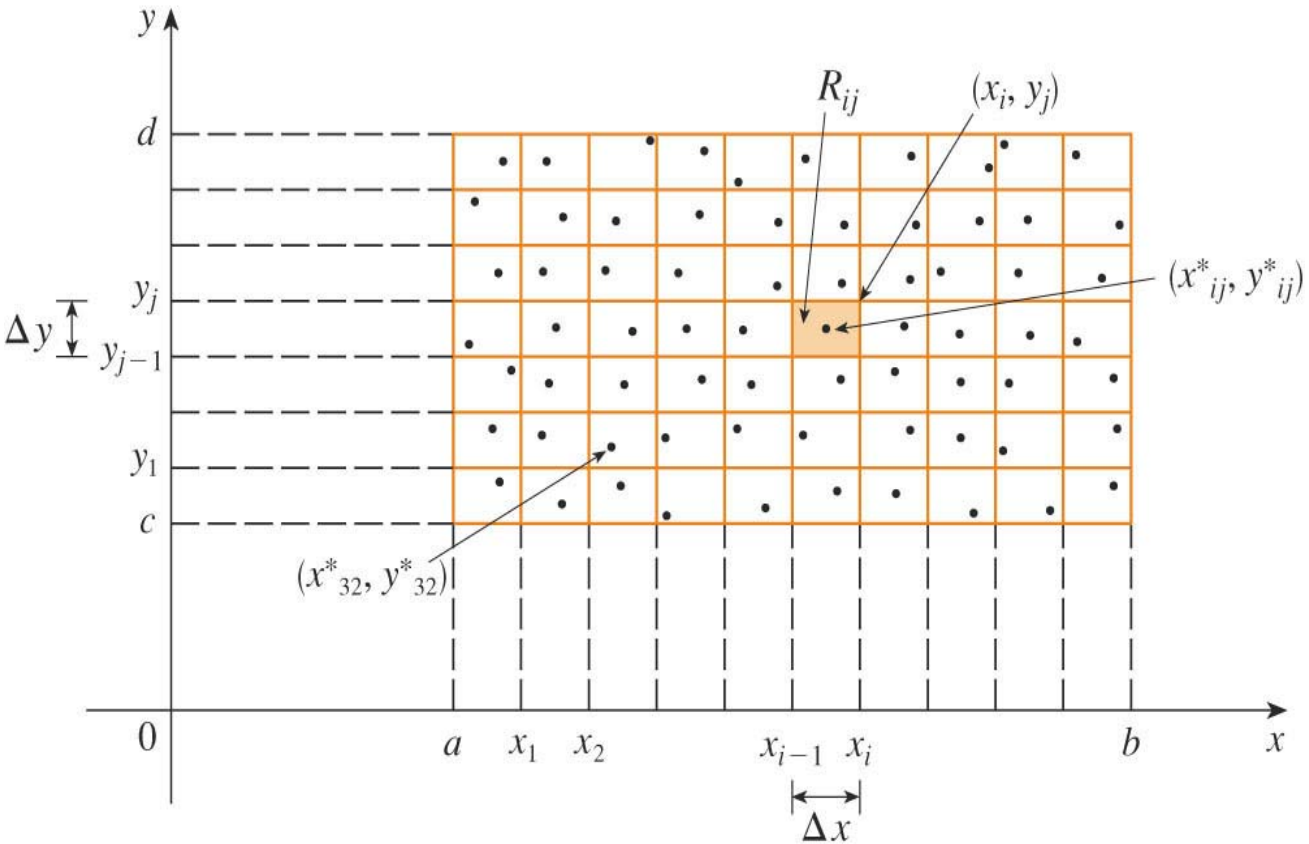
$$R_{ij} = [x_{i-1}, x_i] \times [y_{j-1}, y_j] = \left\{(x, y) : x_{i-1} \leq x \leq x_i,\ y_{j-1} \leq y \leq y_j\right\}$$

for $i = 1, ..., m$ and $j = 1, ..., n$.

The area of R_{ij} is

$$\Delta A_{ij} = (x_i - x_{i-1}) \cdot (y_j - y_{j-1}) = \Delta x \cdot \Delta y.$$

• Diagram: Partitioning rectangle R



• On sub-rectangles

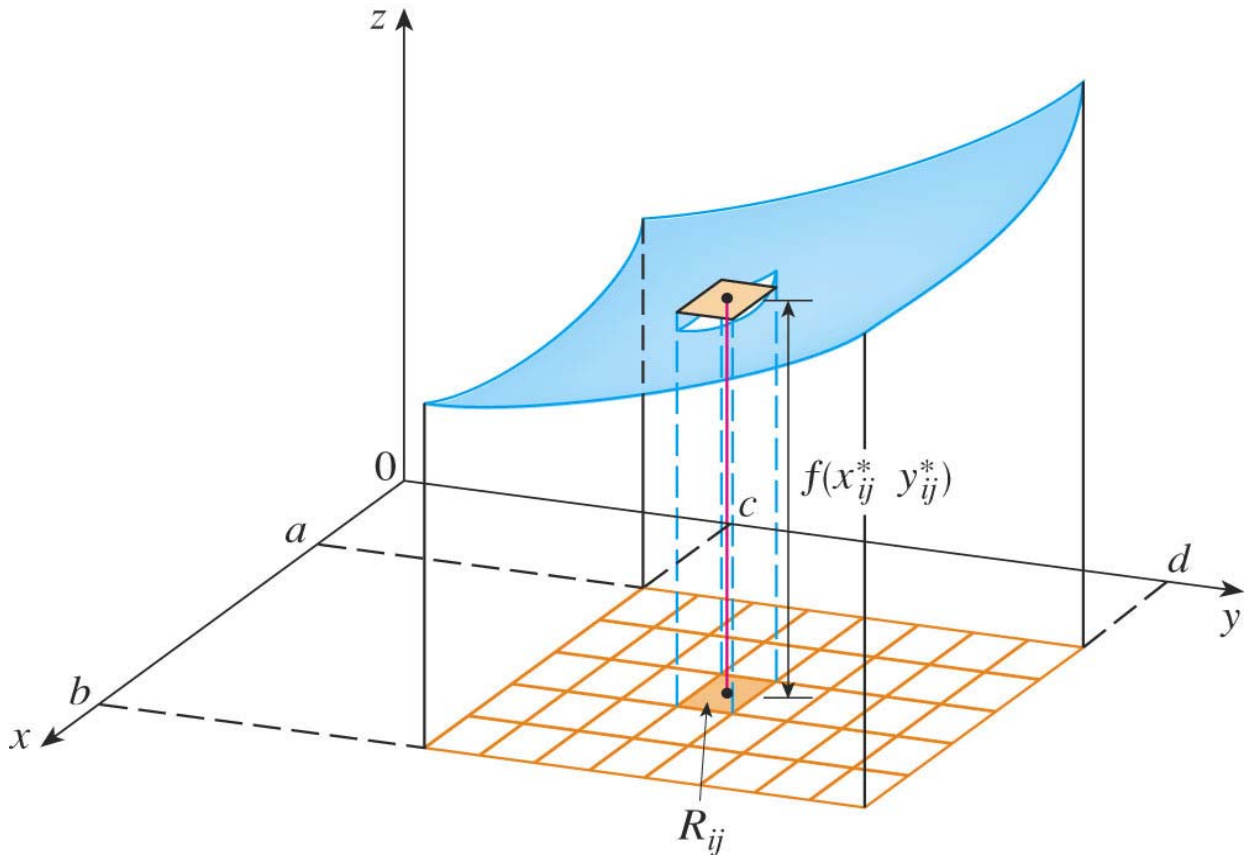
In each sub-rectangle R_{ij}, we choose a point (x_{ij}^*, y_{ij}^*) and compute the value

$$f\left(x_{ij}^*, y_{ij}^*\right) \Delta A_{ij}.$$

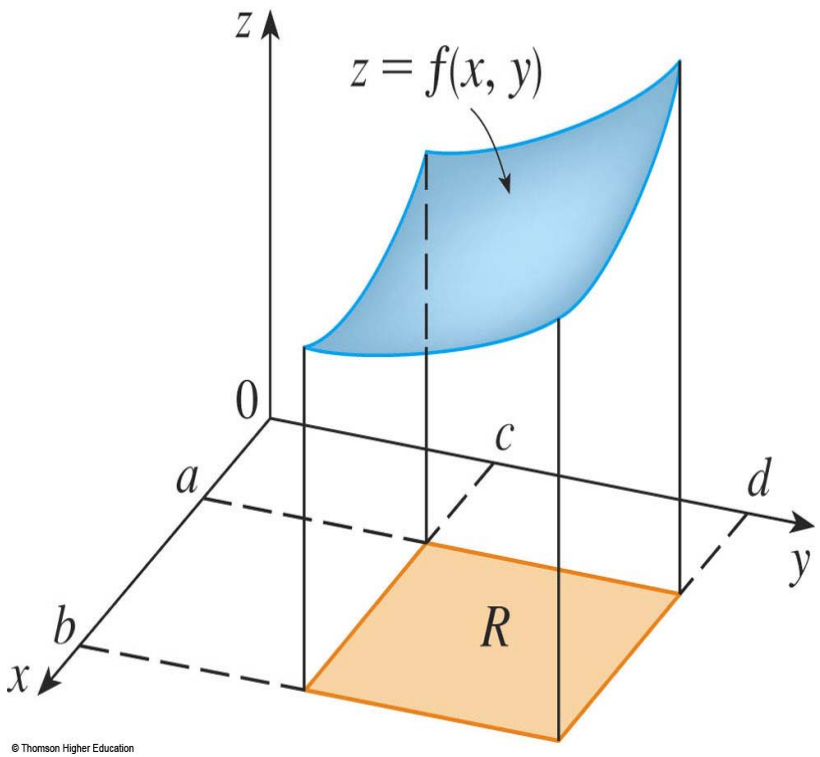
We define the double Riemann sum as

$$\sum_{i=1}^{m} \sum_{j=1}^{n} f(x_{ij}^*, y_{ij}^*) \Delta A_{ij}.$$

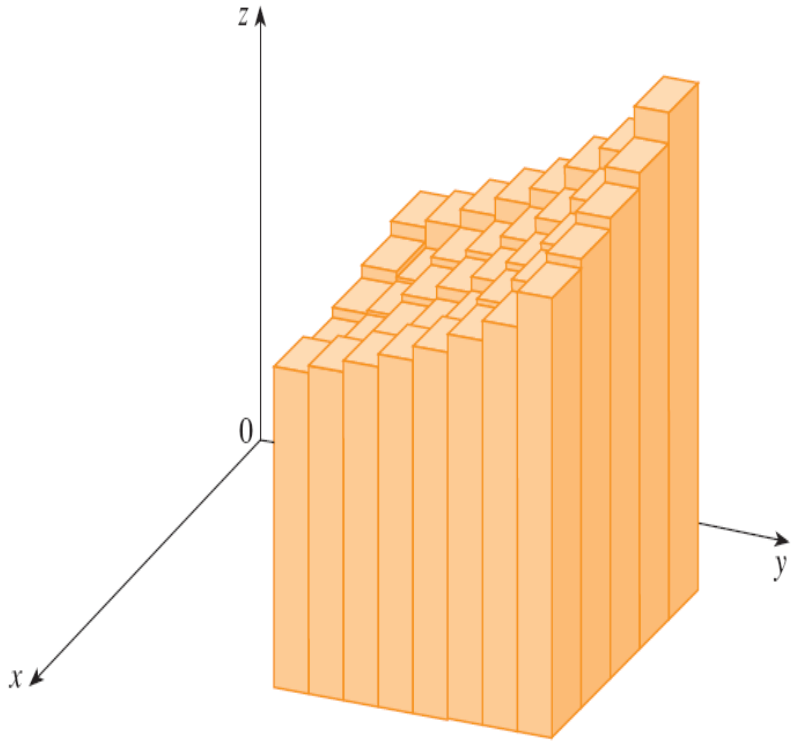
- Diagram: $f(x_{ij}^*, y_{ij}^*)\Delta A_{ij}$



• Diagram: Approximating sums



© Thomson Higher Education



• Double integral over R

Definition

The double integral of $f(x, y)$ over R is defined as follows:

$$\iint_R f(x, y)\, dA = \lim_{\|P\| \to 0} \sum_{i=1}^{m} \sum_{j=1}^{n} f(x_{ij}^*, y_{ij}^*) \Delta A_{ij},$$

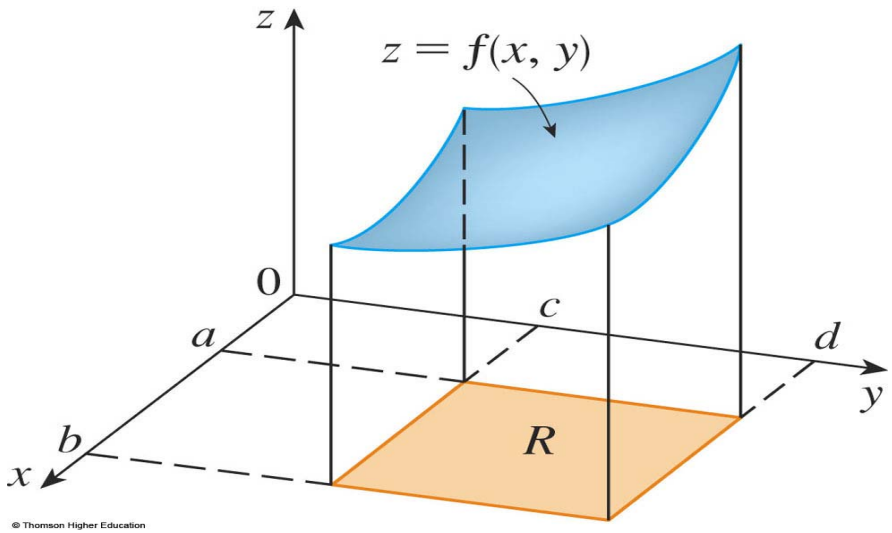
if the limit exists, where $\|P\|$ denotes the longest diagonal of all the sub-rectangles R_{ij}'s, x_{ij}^*, y_{ij}^* and ΔA_{ij} as defined above.

Remark. It is usually difficult to evaluate double integrals by definition. Therefore, we learn how to evaluate such integrals in the sequel.

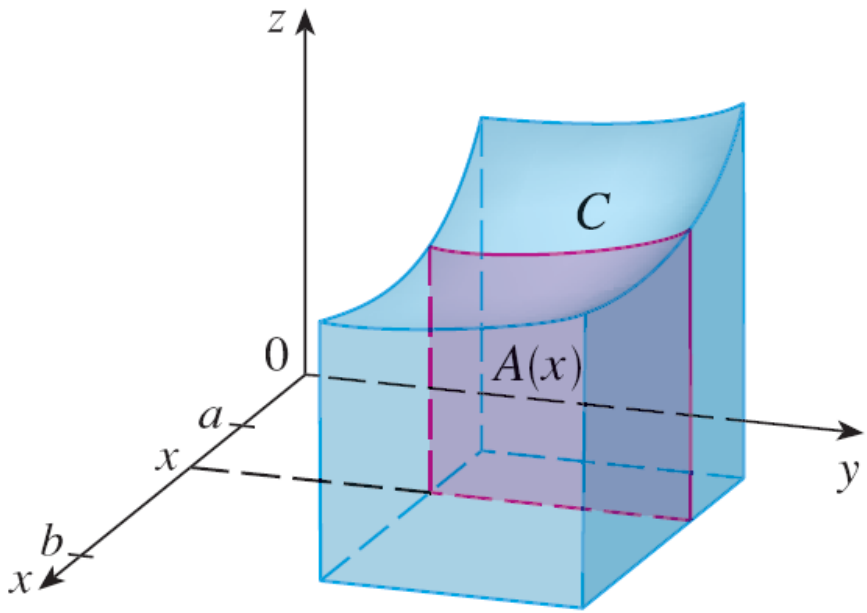
• Iterated Integrals

Now, we see how a double integral can be expressed as an iterated integral, which can be evaluated by calculating two single integrals.

Suppose $f(x, y)$ is a function of two variables that is continuous over the rectangle $R = [a, b] \times [c, d]$.



- **Via slicing, keeping x fixed**



The volume of each thin slice is $A(x) \cdot \delta x$.

For a fixed value of x, we have $A(x) = \int_c^d f(x, y)\, dy$.

Then the volume of each thin slice is $A(x) \cdot \delta x = \left(\int_c^d f(x, y)\, dy \right) \cdot \delta x$.

Summing the volumes of all slices, and allowing the δx to be smaller and smaller, we obtain

$$\int_a^b \left(\int_c^d f(x,y)\, dy \right) dx.$$

- **Iterated integral** $\int_a^b \int_c^d f(x,y)\, dy\, dx$

The iterated integral

$$\int_a^b \int_c^d f(x,y)\, dy\, dx = \int_a^b \left(\int_c^d f(x,y)\, dy \right) dx$$

is obtained by

- first integrate w.r.t. y (holding x fixed) from $y = c$ to $y = d$,
- then integrate the resulting function w.r.t. x from $x = a$ to $x = b$.

- **Via slicing, keeping y fixed**

Similarly, we can also have compute a double integral via slicing, keeping y fixed, to get the iterated integral.

- **Iterated integral** $\int_c^d \int_a^b f(x,y)\,dx\,dy$

The iterated integral

$$\int_c^d \int_a^b f(x,y)\,dx\,dy = \int_c^d \left(\int_a^b f(x,y)\,dx \right) dy$$

is obtained by

- first integrate w.r.t. x (holding y fixed) from $x = a$ to $x = b$,
- then integrate the resulting function w.r.t. y from $y = c$ to $y = d$.

• Evaluating $\iint_R f(x,y)\,dA$

Both $\int_a^b \int_c^d f(x,y)\,dy\,dx$ and $\int_c^d \int_a^b f(x,y)\,dx\,dy$ are iterated integrals (but w.r.t. different orders of x and y).

We have the following important result.

Theorem (5. Fubini's Theorem)

If f is continuous on the rectangle

$R = [a,b] \times [c,d] = \{(x,y) \in \mathbb{R}^2 : a \le x \le b,\ c \le y \le d\}$, *then*

$$\iint_R f(x,y)\,dA = \int_a^b \int_c^d f(x,y)\,dy\,dx = \int_c^d \int_a^b f(x,y)\,dx\,dy.$$

Example

Evaluate the double integral $\iint_R (x - 3y^2)\, dA$, where $R = [0, 2] \times [1, 2]$.

Solution. By Fubini's Theorem, we have two ways to do:

- Integrating w.r.t. y first:

$$\begin{aligned}\iint_R (x - 3y^2)\, dA &= \int_0^2 \int_1^2 (x - 3y^2)\, dy\, dx = \int_0^2 \left[xy - y^3\right]_1^2 dx \\ &= \int_0^2 (x - 7)\, dx = \left[\frac{x^2}{2} - 7x\right]_0^2 = -12.\end{aligned}$$

- Integrating w.r.t. x first:

$$\begin{aligned}\iint_R (x - 3y^2)\, dA &= \int_1^2 \int_0^2 (x - 3y^2)\, dx\, dy = \int_1^2 \left[\frac{x^2}{2} - 3y^2 x\right]_0^2 dy \\ &= \int_1^2 (2 - 6y^2)\, dy = \left[2y - 2y^3\right]_1^2 = -12.\end{aligned}$$

Example

Evaluate $\iint_R y \sin(xy)\, dA$, where $R = [1,2] \times [0,\pi]$.

Solution. By Fubini's Theorem, we may evaluate the double integral as follows (w.r.t. x first):

$$\begin{aligned}\iint_R y \sin(xy)\, dA &= \int_0^{\pi} \int_1^2 y \sin(xy)\, dx\, dy = \int_0^{\pi} \Big[-\cos(xy) \Big]_{x=1}^{x=2} dy \\ &= \int_0^{\pi} [\cos y - \cos(2y)]\, dy = \left[\sin y - \frac{\sin(2y)}{2} \right]_0^{\pi} = 0.\end{aligned}$$

Example (another solution)

- **Question:** How about the other order of integration (i.e. integrate w.r.t. y first)?

We have

$$\iint_R y\sin(xy)\,dA = \int_1^2 \int_0^\pi y\sin(xy)\,dy\,dx.$$

In this case, we should use integration by parts for the internal integral:

$$\int_0^\pi \underbrace{y}_{u}\underbrace{\sin(xy)}_{v'}\,dy = \left[\underbrace{y}_{u}\underbrace{\left(\frac{-\cos(xy)}{x}\right)}_{v}\right]_{y=0}^{y=\pi} - \int_0^\pi \underbrace{1}_{u'}\underbrace{\left(\frac{-\cos(xy)}{x}\right)}_{v}\,dy$$

$$= \frac{-\pi\cos(\pi x)}{x} + \left[\frac{\sin(xy)}{x^2}\right]_{y=0}^{y=\pi} = \frac{1}{x^2}\sin\pi x - \frac{\pi}{x}\cos\pi x.$$

Hence,

$$\iint_R y \sin(xy)\, dA = \int_1^2 \left(\frac{1}{x^2} \sin \pi x - \frac{\pi}{x} \cos \pi x \right)\, dx.$$

Now, observe that

$$\frac{1}{x^2} \sin \pi x - \frac{\pi}{x} \cos \pi x = \frac{d}{dx} \left(\frac{-\sin \pi x}{x} \right).$$

Therefore, we have

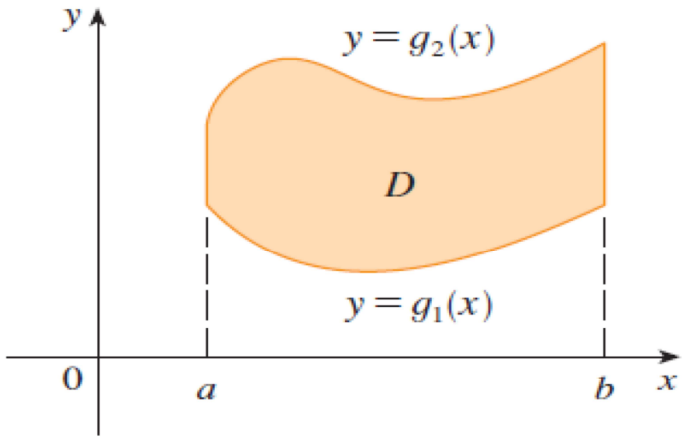
$$\iint_R y \sin(xy)\, dA = \left[-\frac{\sin \pi x}{x} \right]_1^2 = -\frac{\sin 2\pi}{2} + \frac{\sin \pi}{1} = 0.$$

This way is clearly more complicated!

• Double integrals over general regions - I

• Non-rectangular region D of type I

Consider the region D in the sketch:



- The values of x run from a to b.
- For each fixed x, the values of y run from $g_1(x)$ to $g_2(x)$ (on the boundaries, y can be expressed in terms of x).

We describe D as

$$D = \Big\{(x, y) : a \leq x \leq b, \quad g_1(x) \leq y \leq g_2(x)\Big\}.$$

Theorem (8.)

If f is continuous on a region D of type I, i.e. on

$$D = \Big\{(x, y) : a \leq x \leq b, \quad g_1(x) \leq y \leq g_2(x)\Big\},$$

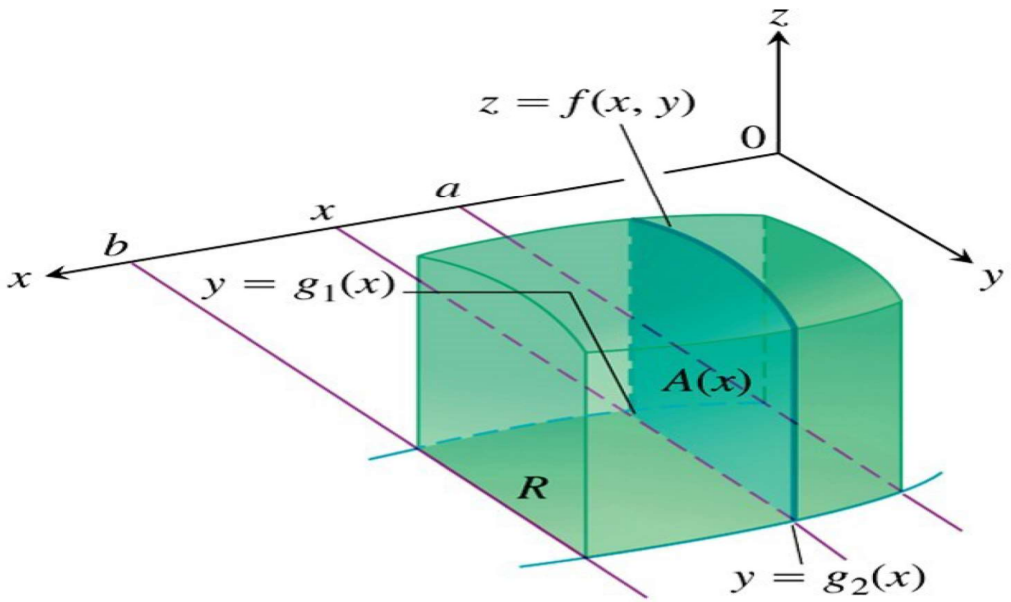
then

$$\iint_D f(x, y)\, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\, dy\, dx.$$

• Diagram: Type I

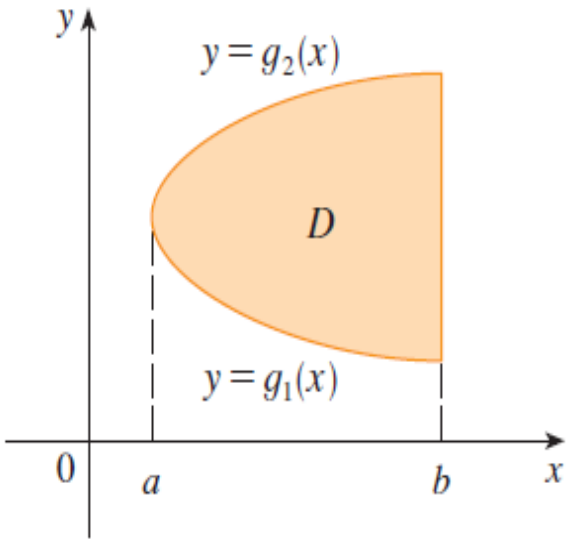
The area of the vertical slice shown here is $A(x)$. To calculate the volume of the solid, we integrate this area from $x = a$ to $x = b$:

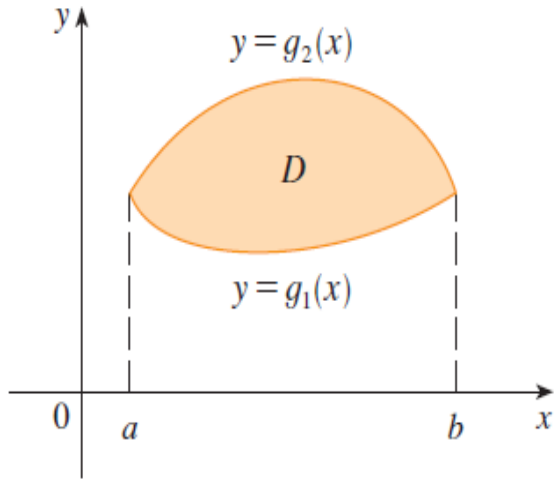
$$\int_a^b A(x)\,dx = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)\,dy\,dx.$$



• Double integrals over general regions - I

- Some other non-rectangular regions D of type I



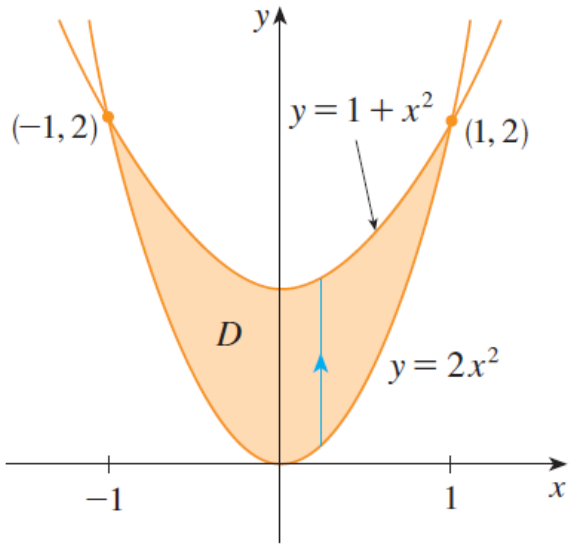


Example

Evaluate $\iint_D (x + 2y)\, dA$, where D is the region bounded by the parabolas $y = 2x^2$ and $y = 1 + x^2$.

Solution. *It is useful to sketch the region D.*

Note that the two parabolas intersect when $2x^2 = 1 + x^2 \iff x = \pm 1$.



From the sketch, we have the values of x run from -1 to 1. For each fixed x, the values of y run from $2x^2$ to $1 + x^2$. Thus

$$D = \left\{(x, y) : -1 \leq x \leq 1, \quad 2x^2 \leq y \leq 1 + x^2\right\}.$$

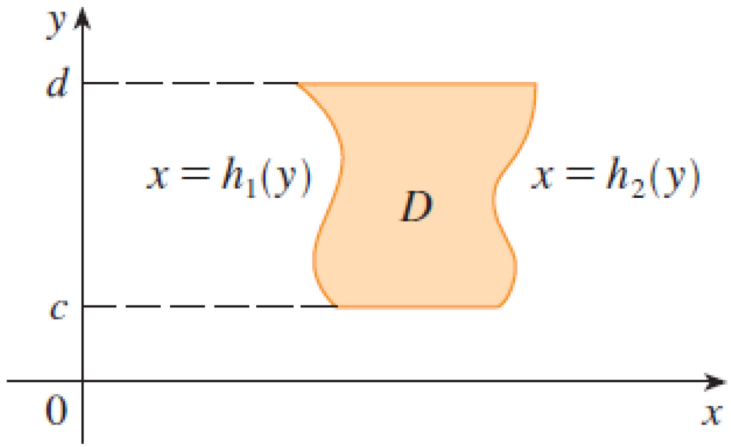
Then we have

$$\begin{aligned}
\iint_D (x + 2y)\, dA &= \int_{-1}^{1} \int_{2x^2}^{1+x^2} (x + 2y)\, dy\, dx \\
&= \int_{-1}^{1} \left[xy + y^2\right]_{y=2x^2}^{y=1+x^2} dx \\
&= \int_{-1}^{1} \left(-3x^4 - x^3 + 2x^2 + x + 1\right) dx = \frac{32}{15}.
\end{aligned}$$

• Double integrals over general regions - II

• Non-rectangular region D of type II

Consider the region D in the sketch:



- The values of y run from c to d.
- For each fixed y, the values of x run from $h_1(y)$ to $h_2(y)$ (on the boundaries, x can be expressed in terms of y).

So we have

$$D = \Big\{(x, y) : c \leq y \leq d, \quad h_1(y) \leq x \leq h_2(y)\Big\}.$$

Similarly, we have the following result.

Theorem

If f is continuous on a region D of type II, i.e. on

$$D = \Big\{(x, y) : c \leq y \leq d, \quad h_1(y) \leq x \leq h_2(y)\Big\},$$

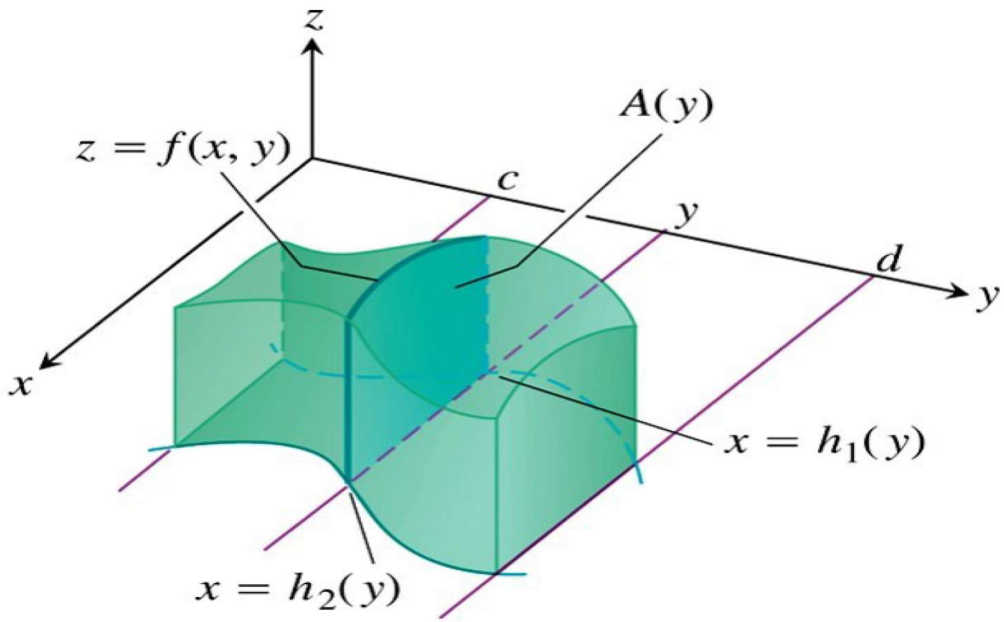
then

$$\iint_D f(x, y)\, dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y)\, dx\, dy.$$

• Diagram: Type II

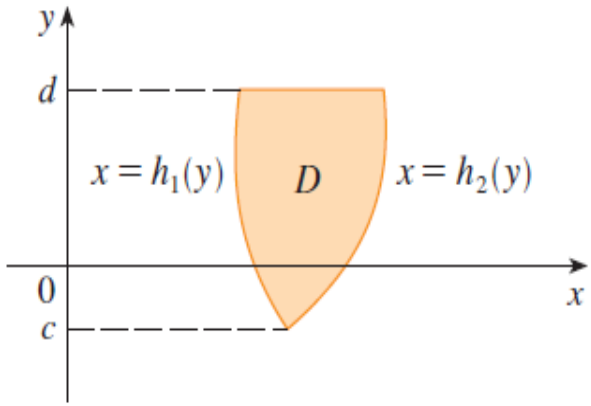
The area of the vertical slice shown here is $A(x)$. To calculate the volume of the solid, we integrate this area from $y = c$ to $y = d$:

$$\int_c^d A(y)\,dy = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,y)\,dx\,dy.$$



• Double integrals over general regions - II

- Some other non-rectangular regions D of type II



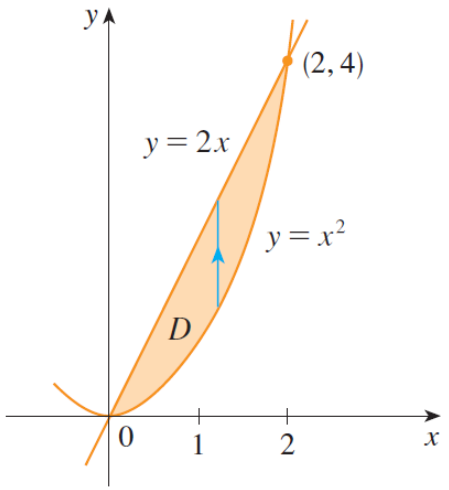
Example

Find the volume of the solid that lies under the paraboloid $z = x^2 + y^2$ and above the region D in the xy-plane bounded by the line $y = 2x$ and the parabola $y = x^2$.

Solution. Note that the volume under $z = x^2 + y^2$ and above D is

$$V = \iint_D (x^2 + y^2)\, dA.$$

The two parabolas intersect when $2x = x^2 \iff x = 0,\ x = 2$.



- First, we can describe D as a region of type I:

$$D = \left\{(x, y) : 0 \leq x \leq 2, \quad x^2 \leq y \leq 2x\right\}.$$

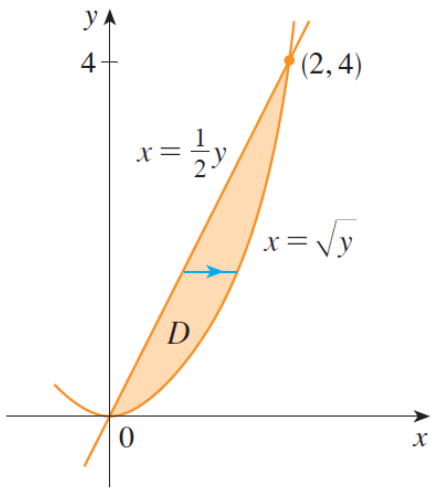
Then the volume under $z = x^2 + y^2$ and above D is

$$\begin{aligned} V &= \iint_D (x^2 + y^2)\, dA = \int_0^2 \int_{x^2}^{2x} (x^2 + y^2)\, dy\, dx \\ &= \int_0^2 \left(\left[x^2 y + \frac{1}{3} y^3 \right]_{y=x^2}^{y=2x} \right) dx \\ &= \int_0^2 -\frac{1}{3} x^3 (x^3 + 3x - 14)\, dx = \frac{216}{35}. \end{aligned}$$

Example (Alternative solution)

- We can also describe D as a region of type II:

$$D = \left\{(x, y) : 0 \leq y \leq 4, \quad \frac{y}{2} \leq x \leq \sqrt{y}\right\}.$$



Example (Alternative solution - cont.)

Then, we have

$$\begin{aligned} V &= \iint_D (x^2+y^2)\,dA = \int_0^4 \int_{y/2}^{\sqrt{y}} (x^2+y^2)\,dx\,dy \\ &= \int_0^4 \left(\frac{1}{3}y^{\frac{3}{2}} - \frac{13}{24}y^3 + y^{\frac{5}{2}}\right)\,dy = \frac{216}{35}. \end{aligned}$$

Example

Evaluate $\iint_D xy\, dA$, where D is the region bounded by the line $y = x - 1$ and the parabola $y^2 = 2x + 6$.

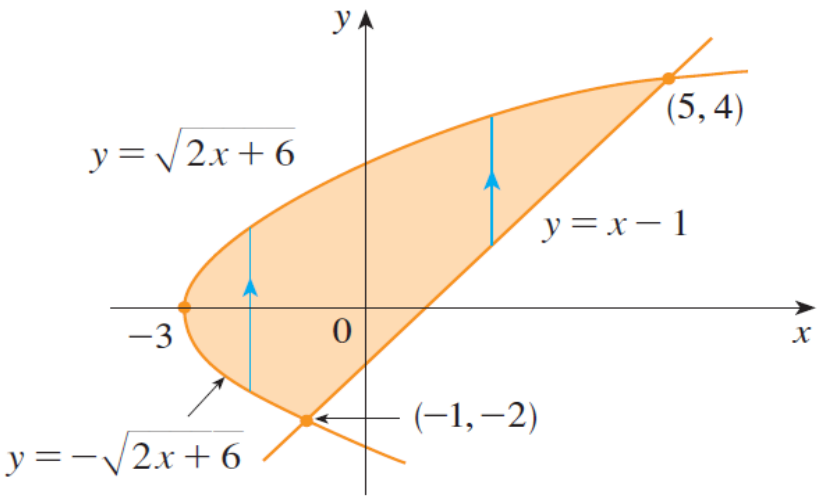
Solution. Points of intersection are

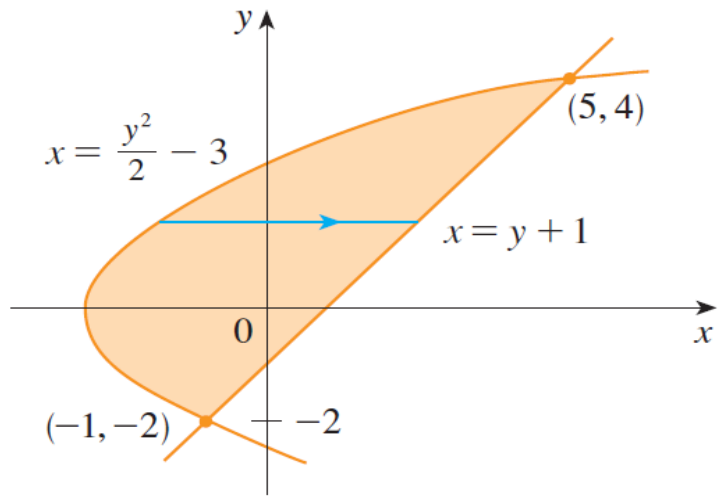
$$(x-1)^2 = 2x + 6 \iff x^2 - 4x - 5 = 0$$

$$\iff (x-5)(x+1) = 0 \iff x = -1,\ x = 5.$$

Example (cont.)

We can describe D as a region of either type I (the left figure) or type II (the right figure):





Example (cont.)

If we use a description of D as a region of type II, that is

$$D = \left\{(x, y) : -2 \leq y \leq 4, \quad \frac{y^2}{2} - 3 \leq x \leq y + 1\right\},$$

then we have

$$\begin{aligned}\iint_D xy \, dA &= \int_{-2}^{4} \int_{\frac{y^2}{2}-3}^{y+1} xy \, dx \, dy = \int_{-2}^{4} \left[\frac{x^2 y}{2}\right]_{x=\frac{y^2}{2}-3}^{x=y+1} dy \\ &= \int_{-2}^{4} \frac{1}{8} y(-y^4 + 16y^2 + 8y - 32) \, dy = 36.\end{aligned}$$

If we would have used a description of D as a region of type I, then we would have more work involved - we come back to this later.

Example (Changing the order of integration)

Evaluate the iterated integral

$$\int_0^1 \int_x^1 \sin(y^2)\, dy\, dx.$$

Solution. If we wish to integrate w.r.t. y first, then we have to evaluate $\int \sin(y^2)\, dy$. Unfortunately, it is impossible to find the integral of $\sin(y^2)$ as it is not an elementary function.

The trick we apply here is to change the order of integration, i.e., we evaluate the integral $\int_0^1 \int_x^1 \sin(y^2)\, dy\, dx$ by $\int_?^? \int_?^? \sin(y^2)\, dx\, dy$.

How to do?

Example (cont.)

Technique: Sketch the region D and re-describe it.

From the iterated integral $\int_0^1 \int_x^1 \sin(y^2)\, dy\, dx$, we note that

$$0 \leq x \leq 1, \quad x \leq y \leq 1.$$

Then D is the region

$$D = \Big\{(x, y) : 0 \leq x \leq 1, \quad x \leq y \leq 1\Big\},$$

which is a region of type I (which requires to integrate w.r.t. y first).

Now we sketch D and describe it as the region of type II:

$$D = \Big\{(x, y) : 0 \leq y \leq 1, \quad 0 \leq x \leq y\Big\}.$$

So we evaluate the given integral by changing the order:

$$\int_0^1 \int_x^1 \sin(y^2)\, dy\, dx = \int_0^1 \int_0^y \sin(y^2)\, dx\, dy.$$

Example 13 (cont.)

We have

$$\begin{aligned}\int_0^1 \int_0^y \sin(y^2)\, dx\, dy &= \int_0^1 \Big[x \sin(y^2)\Big]_{x=0}^{x=y} dy \\ &= \int_0^1 y \sin(y^2)\, dy \\ &= \left[-\frac{1}{2}\cos(y^2)\right]_0^1 = \frac{1}{2}(1-\cos 1).\end{aligned}$$

Therefore,

$$\int_0^1 \int_x^1 \sin(y^2)\, dy\, dx = \frac{1}{2}(1-\cos 1).$$

Properties of double integrals

- **Over Non-overlapping region**

Proposition

If $R = R_1 \cup R_2$, where R_1 and R_2 are two non-overlapping regions, then

$$\iint_R f(x,y)\,dA = \iint_{R_1} f(x,y)\,dA + \iint_{R_2} f(x,y)\,dA.$$

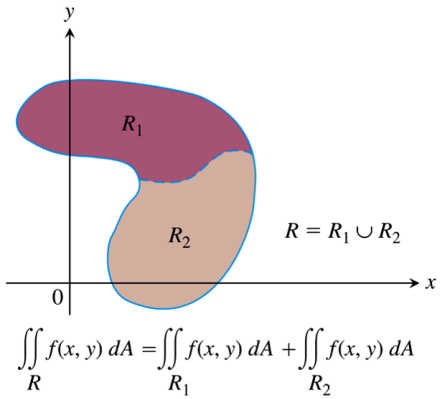
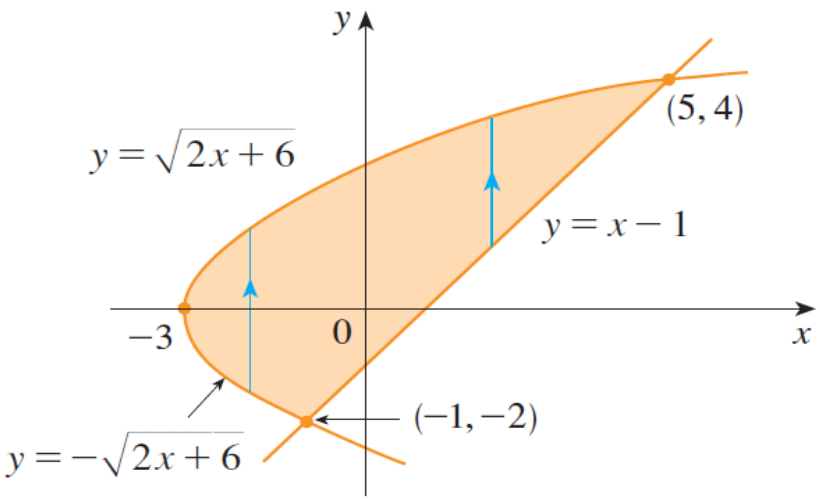


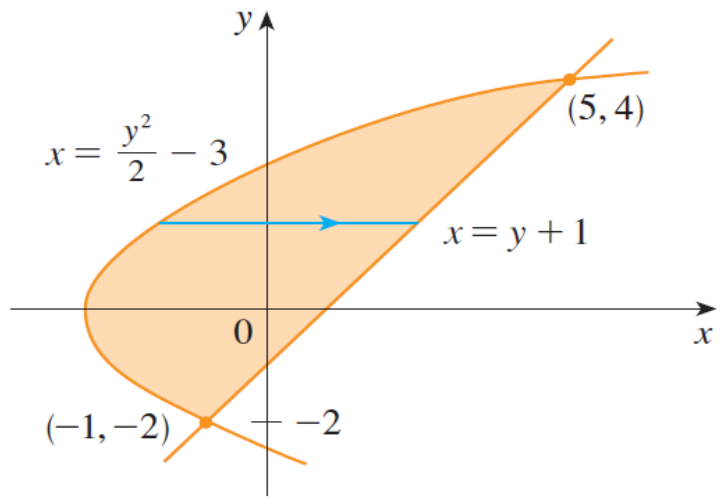
FIGURE 15.17 The Additivity Property for rectangular regions holds for regions bounded by continuous curves.

Example (Come back to one of examples.)

Evaluate $\iint_D xy\, dA$, where D is the region bounded by the line $y = x - 1$ and the parabola $y^2 = 2x + 6$.

We already know two descriptions of D, as a region of either type I (the left figure) or type II (the right figure)





Example - cont.

If we use a description of D as a region of type I, then D is the union of two non-overlapping regions $D_1 \cup D_2$, where

$$D_1 = \left\{(x, y) : -3 \leq x \leq -1, -\sqrt{2x+6} \leq y \leq \sqrt{2x+6}\right\}$$

and

$$D_2 = \left\{(x, y) : -1 \leq x \leq 5, x - 1 \leq y \leq \sqrt{2x+6}\right\}.$$

In this case, by Proposition 14, we have

$$\iint_D xy \, dA = \int_{-3}^{-1} \int_{-\sqrt{2x+6}}^{\sqrt{2x+6}} xy \, dy \, dx + \int_{-1}^{5} \int_{x-1}^{\sqrt{2x+6}} xy \, dy \, dx.$$

It is clear that this needs more work than the first method.

• Linear Property of Double Integrals

Proposition

Suppose

- *f and g are functions of two variables defined on D,*
- *f and g are both integrable over D.*

Then, for any real numbers c and d, we have

$$\iint_D [c\, f(x,y) + d\, g(x,y)]\; dA = c \iint_D f(x,y)\; dA + d \iint_D g(x,y)\; dA.$$

• Order Properties of Double Integrals

Proposition

(a) *If* $f(x,y) \geq g(x,y)$ *for all* $(x,y) \in D$, *then*

$$\iint_D f(x,y)\, dA \geq \iint_D g(x,y)\, dA.$$

(b) *If* $m \leq f(x,y) \leq M$ *for all* $(x,y) \in D$, *then*

$$\underbrace{\iint_D m\, dA}_{m\,A(D)} \leq \iint_D f(x,y)\, dA \leq \underbrace{\iint_D M\, dA}_{M\,A(D)}.$$

END OF CHAPTER 2